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Pest-PVT: A model for multi-class and dense pest detection and counting in field-scale environments

Hongrui Chen [a,b], Changji Wen [a,b,*,1], Long Zhang [a,b], Zhenyu Ma [a,b], Tianyu Liu [a,b], Guangyao Wang [a,b], Helong Yu [a,b], Ce Yang [b,c], Xiaohui Yuan [b,d], Junfeng Ren [a,b,*,1]

[a] College of Information and Technology, Jilin Agricultural University, Changchun, China
[b] Institute for the Smart Agriculture, Jilin Agricultural University, Changchun, China
[c] College of Food, Agricultural and Natural Resource Sciences, University of Minnesota, Twin Cities, MN, United States
[d] Edible and Medicinal Fungi Engineering Research Center of the Ministry of Education, Jilin Agricultural University, Changchun, China

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ABSTRACT

Field-scale pest monitoring is crucial for evaluating insect infestations in agricultural environments. The Pest24 dataset presents unique challenges due to the small pest target sizes, high target similarity, and dense pest distribution. To address these challenges, we propose Pest-PVT, a comprehensive framework based on the Pyramid Vision Transformer v2 (PVTv2) network model. Pest-PVT adopts an anchor-free approach using Fully Convolutional One-Stage Object Detection (FCOS) to enhance small object detection and incorporates Adaptive Training Sample Selection (ATSS) to mitigate sample imbalance bias. Dynamic Heads (DyHead) are employed to handle pests of varying scales and spatial changes, while Shunted Self-Attention (SSA) enhances multi-scale feature capture and reduces memory consumption. Compared to 23 other mainstream detection models and previously published works, our proposed Pest-PVT outperforms state-of-the-art detectors on pest detection datasets, achieving the outperformed scores in detection evaluation metrics such as mAP, Precision, Recall, and F1-Score, reaching 77.2 %, 78.42 %, 81.27 %, and 0.80, respectively. With a parameter size of only 24.74 M, Pest-PVT is suitable for integration into edge devices with limited resources. This work represents a significant contribution to the field of field-scale pest monitoring and addresses the specific challenges posed by the Pest24 dataset. Our code is made available at https://github.com/jlauwcj/pest-pvt.

1. Introduction

Field-scale pest monitoring is a crucial method for monitoring insect infestations. It is an important tool to assess the severity of pest damage and to help farmers make informed decisions about pest control measures. By closely observing pest populations in the field, farmers can detect early signs of infestation and take appropriate action before significant crop damage occurs. Effective field-scale pest monitoring requires careful planning, accurate data collection, and timely analysis of the results (Way and Van Emden, 2000). This method can serve as a valuable and essential supplement to large-scale insect infestation monitoring techniques, including high-altitude remote sensing. Due to the diverse types and varied morphology of pests, as well as their rapid and severe damage potential, it was crucial to achieve automated, rapid, and accurate detection and identification models of pest populations during the early stages. This is essential for predicting the size and distribution of pest populations in the field, and for implementing effective pest control measures (Høye et al., 2021).

In recent years, the development of computer vision and deep learning technology has provided new solutions for automatic pest identification. R. Li et al. (2019) proposed a two-stage network for aphid recognition and detection. This network employs a coarse-to-fine detection pipeline, initially leveraging an Improved Non-Maximum Suppression (INMS) strategy to select potential target regions within naturally occurring images featuring dense clusters of aphids. Subsequently, the refined detection network processes these candidate regions to enhance detection accuracy. Experimental data demonstrate that this method achieves an Average Precision (AP) of 76.8 %. He et al. (2020) proposed a two-layer detection algorithm based on deep learning technology for rapidly and accurately detecting the major pest of rice,

* Corresponding authors at: College of Information and Technology, Jilin Agricultural University, Changchun, Jilin Province 130118, China.
E-mail addresses: changjiw@jlau.edu.cn (C. Wen), renjunfeng@mails.jlau.edu.cn (J. Ren).

[1] Both authors have equally contributed, and should be considered co-first corresponding author.

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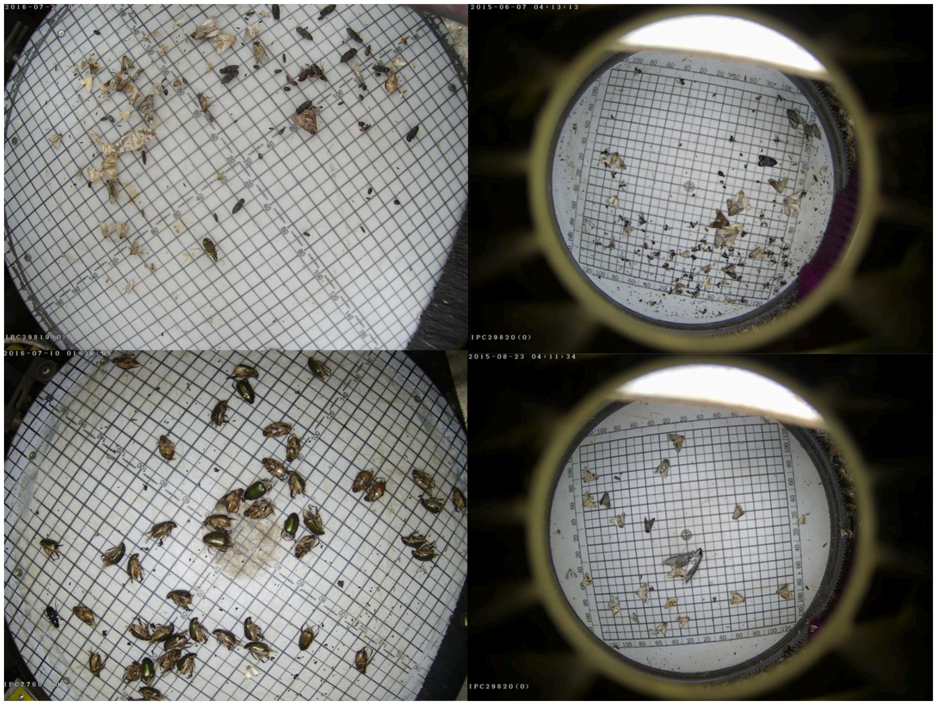

Fig. 1. The partial example images from the pest24.

Nilaparvata lugens Stal. Due to the small size, large number, and complex background of brown planthoppers, it was challenging to detect them using images. The research adopted a two-layer detection structure, in which each layer used the Faster RCNN (with convolutional neural network feature regions) algorithm, and selected different feature extraction networks for different layers to effectively utilize computational resources. Experimental data showed that the two-layer detection algorithm had higher mAP and Recall than YOLOv3 in detecting brown planthoppers. F. Wang et al. (2020a) proposed a two-stage cascade detection method called DeepPest, which consists of two stages. First, a context-aware attention network was used to classify the original image, and then an optimized multi-projection pest detection model was applied to capture fine pest features. In practical applications, compared to existing algorithms such as VGG-16 + FPN and ResNet-50 + FPN, DeepPest had an advantage in mAP of 73.9 % when detecting small pests densely distributed in the wild. Du et al. (2022) proposed a detection task called Densely Clustered Tiny (DCT) and constructed a new dataset APHID-4 K for this task, which includes samples of two aphids that often appear in clusters in real agricultural environments. To tackle the challenges posed by the DCT detection task, researchers devise a novel network architecture, DCTDet, which leverages density information to enhance the performance in detecting such densely clustered tiny objects. Empirical findings reveal that the proposed DCTDet method achieves an AP of 33.9 % on the APHID-4 K dataset. Chodey and Noorullah Shariff (2022) proposes a hybrid deep learning model for real-time monitoring and accurate identification of pests in farmland. This model was designed to address the issue of growth retardation and yield decline in important crops caused by pest infestation, overcoming the limitations of traditional methods that struggle to balance execution speed and calculation cost. Experimental results demonstrate the effectiveness of the proposed method on various reference datasets, with a measured structural similarity index (SSIM) value reaching 0.99, mean absolute error (MAE) less than 0.2, and AP reaching 89.67 %

With the growing adoption of deep learning models in agricultural pest detection, the availability of large-scale pest data and dependable data annotation has emerged as a bottleneck hindering the progress of pest detection tasks in a real-world agricultural production environment. Among the currently available large-scale pest datasets, the Pest24 stood as one of the most exemplary datasets. The Pest24 dataset contains 24 categories of major pests of field crops designated by the Ministry of Agriculture of China as needing focused monitoring. And these 24 pests caused significant damage to crops every year in China and belong to typical agricultural pest categories (Q.-J. Wang et al., 2020). Fig. 1 shows the Partial Example Images from the Pest24, which is a large multi-pest image dataset that poses unique challenges such as small pest target sizes, high target similarity, and dense pest distribution. The detailed characteristics analysis of the Pest24 dataset can be found in the 2.2 section, Data Analysis. Therefore, the models trained on this dataset were better suited to the practical needs of pest management, making them more challenging and useful for real-world applications.

Table 1 provides an overview of the relevant studies conducted in the past few years on the Pest24 dataset. Based on the experimental results presented in Table 1, it is evident that achieving accurate pest detection on Pest24 remains an exceedingly challenging task. The aforementioned work demonstrated that improved detection algorithms and feature extraction techniques can achieve efficient pest detection under specific conditions. However, the accuracy and robustness of existing methods still need improvement when dealing with extremely small or densely distributed targets. Additionally, the morphological similarity between different pest species increases classification difficulty, especially in poor-quality images where subtle features are hard to distinguish.

In response to the aforementioned challenges, we propose a comprehensive model framework, known as Pest-PVT, built upon the Pyramid Vision Transformer v2 network model. PVTv2 (W. Wang et al., 2022) stood out in dense small object detection due to its enhanced multi-scale features, efficient computation, robustness in crowded scenes, and state-of-the-art performance. Firstly, in the field of dense small object detection, if the scales and aspect ratios of the pre-set anchor boxes are not appropriately chosen to capture the distinctive features of these objects, it could hinder the effective detection of such objects. As a result, this issue could potentially erode the accuracy of the overall object detection process (Lv et al., 2022). To reduce the sensitivity of pre-set anchor boxes, we use an anchor-free detection framework Fully Convolutional One-Stage Object Detection (FCOS) (Tian et al., 2019). Secondly, severe sample imbalance, such as imbalance in

Table 1
Related work on the Pest24 dataset.

Model	Num of Classes	mAP	Recall	Flops
AF-RCNN[a] (Jiao et al., 2020)	24	56.42 %	85.10 %	/
Pest-YOLO[b] (Tang et al., 2021)	24	71.6 %	83.5 %	/
YOLOv3-GCNet[c] (Q. Liu, Yan, Wang, & Ding, 2021)	24	65.69 %	/	/
Improved YOLOv3[d] (Lv et al., 2022)	9[e]	77.28 %	68.90 %	/
YOLOv5 (Teixeira et al. (2022)	3[f]	95.5 %	/	/
Faster R-CNN-TSDg (Huang et al., 2022)	24	51.0 %	/	/
Improved deformable DETR[h] (Qi et al., 2022)	24	72.5 %	/	170G
Pest-YOLO[i] (Wen et al., 2022)	24	69.59 %	77.71 %	30.09G
Improved YOLOv5[j] (Teixeira et al., 2023)	24	72.1 %	/	/
BF-YOLO[k](Qi et al., 2023)	24	74.1 %	/	66.3G

[a] Design of a novel anchor-free regional proposal generation network (AFRPN) that can be applied to two-stage pest detection
[b] Introducing the Squeeze-and-Excitation (SE) attention module to CNN.
[c] Using Improved GC-Net for Feature Fusion.
[d] Anchor boxes were optimised using a k-means based linear transformation method and the Draknet53 backbone network was improved.
[e] selected nine common species of maize pest, namely Armyworm, Bollworm, Athetis Lepigone, Little Gecko, Yellow Tiger, Holotrichia Oblita, Holotrichia Parallela, Anomala Corpulenta, Agriotes Fuscicollis Miwa.
[f] The Pest24 dataset was reorganised into a dataset with three classes according to the pest samples with the highest numbers, highest colour differences and largest relative proportions. [g] Proposed a truncated structurally-aware loss that directly considers different geometric relationships between adjacent bounding boxes.
[h] Use of a multi-head cross-attention module to replace the self-attention module in a deformable DETR.
[i] Use improved I-confidence loss and confluence candidate box selection strategies.
[j] Tuning the hyperparameters of YOLOv5 using an evolutionary algorithm.
[k] Using GhostNet as the backbone and introducing BiFPN and feature fusion with the attentional multiple receptive fields (FFARF) to weight the multi-scale receptive fields.

the distribution of pest species, as well as an imbalance between pest and background classes, leads to model bias, thereby reducing both accuracy and generalization capability. That is a common problem in the field of agriculture that has a significant impact on model training. To address this challenging issue, we introduce a sample re-balancing strategy, Adaptive Training Sample Selection (ATSS) (Zhang et al., 2020), which dynamically selects training samples that better match the distribution of pest samples for model training. Thirdly, the diversity of pest categories in the Pest24 dataset, as well as the random distribution of pests within the images, can significantly affect the model's detection performance in a negative way. Therefore, it is crucial for the detection head of the model to effectively handle the variability in target scales and the complexity of spatial relationships. In the neck structure of our model,we have incorporated stacked Dynamic Heads (DyHead) (Dai et al., 2021) to handle pests of different scales, capture spatial changes in pest positions, and adapt to various pest detection task requirements. Finally, the traditional Vision Transformer (ViT) partitions the input image into patches and updates their features using global self-attention. This self-attention mechanism is effective at capturing long-range dependencies and allowing patches to incorporate information from other patches, which has contributed to the success of ViT. However, traditional ViT models face limitations in capturing multi-scale features and exhibit performance degradation when dealing with images containing objects of different scales, like Pest24. Additionally, self-attention incurs high memory consumption costs. To address these challenges, the Shunted Self-Attention (SSA) method (Sucheng (Ren et al., 2022) introduces a selective token merging strategy, enhancing multi-scale feature capture and reducing memory consumption.

The contributions of this work are described as follows.

The Pest24 dataset poses significant challenges in agricultural pest monitoring, such as small sizes, high visual similarity, and dense distribution. To address these, we developed the Pest-PVT framework using Pyramid Vision Transformer v2 (PVTv2) for multi-scale feature extraction and global context capture. We further optimized PVTv2 with the Shunted Self-Attention (SSA) module, which processes coarse- and fine-grained features independently, enhancing multi-scale capabilities and reducing memory usage. This improves adaptability and detection accuracy in complex environments.

To address the small size characteristics of agricultural pests, we utilized an anchor-free Fully Convolutional One-Stage (FCOS) detector combined with Adaptive Training Sample Selection (ATSS). This approach resolves the poor performance of traditional methods when handling extremely small or densely arranged targets, thereby improving detection accuracy. Furthermore, the incorporation of Dynamic Head (DyHead) significantly enhances the model's ability to handle spatial variations across different scales of pests, leading to notably improved detection performance in high-density distributions.

The Pest-PVT proposed in this paper outperforms the state-of-the-art (SOTA) detector on the pest24 detection dataset. Compared to mainstream detection models and 23 other previously published models, Pest-PVT achieves superior performance in detection evaluation metrics such as mAP, precision, recall, and f1-score, which are 77.2 %, 78.42 %, 81.27 %, and 0.80, respectively. In addition, the parameter quantity of Pest-PVT is only 24.74 M, which is very suitable for deployment in edge computing devices and real-time pest detection and counting in resource constrained environments. Pest-PVT not only verified its superior performance in the laboratory, but also demonstrated its potential for wide application in the actual agricultural environment.

2. Materials

2.1. Image data acquisition

The Pest24 dataset is collected by a specialized automatic pest image collection device manufactured by the Institute of Intelligent Machines, Hefei Institutes of Physical Science, Chinese Academy of Sciences. The process of trapping pests and collecting images is as follows. Firstly, a multispectral trap is used to trap various types of pests based on their sensitivity to specific spectral bands. This kind of trap utilizes pest-sensitive wavelengths to attract and capture the pests. Once attracted to the specific wavelengths of the spectrum, the pests descend onto a pest collection plate through an impact blocking plate. Afterwards, they are collected in a pest collection funnel that is connected to the plate. This setup ensures the safe collection of trapped pests, allowing for easy retrieval for further examination or disposal. Utilizing the HD camera loaded on the collection plate to periodically acquire images. After each photo is taken, the pests are regularly removed from the pest collection tray to avoid excessive accumulation and overlapping. The resolution of the images captured by the device is 2095 × 1944 and they are stored in JPG format. The collection personnel manually screened the collected pest images, deleting abnormal data such as lens obstructions and blurred images, resulting in 25,378 images being collected. The captured images include 24 types of crop pests required by the Ministry of Agriculture and Rural Affairs of China to be monitored. The annotation of the dataset was performed using the "lableImg" software by plant

Table 2
Number of images and instances of each type of pest in the Pest24 dataset (Wang et al., 2020).

Index	Pest name	images	instances	Index	Pest name	images	instances
1	Rice planthopper	316	1,511	13	Spodoptera cabbage	1,707	2,302
2	Rice Leaf Roller	944	1,240	14	Scotogramma trifolii Rottemberg	3,223	4,679
3	Chilo suppressalis	454	1,285	15	Yellow tiger	1,388	1,686
4	Armyworm	3,824	8,880	16	Land tiger	369	475
5	Bollworm	9,049	28,014	17	Eight-character tiger	154	168
6	Meadow borer	5,526	16,516	18	Holotrichia oblita	90	108
7	Athetis lepigone	7,520	30,339	19	Holotrichia parallela	3,111	11,675
8	Spodoptera litura	1,588	1,951	20	Anomala corpulenta	5,228	53,347
9	Spodoptera exigua	3,614	7,263	21	Gryllotalpa orientalis	3,629	6,528
10	Stem borer	1,357	1,804	22	Nematode trench	118	167
11	Little Gecko	2,503	4,279	23	Agriotes fuscicollis Miwa	1,814	6,484
12	Plutella xylostella	531	953	24	Melahotus	239	768

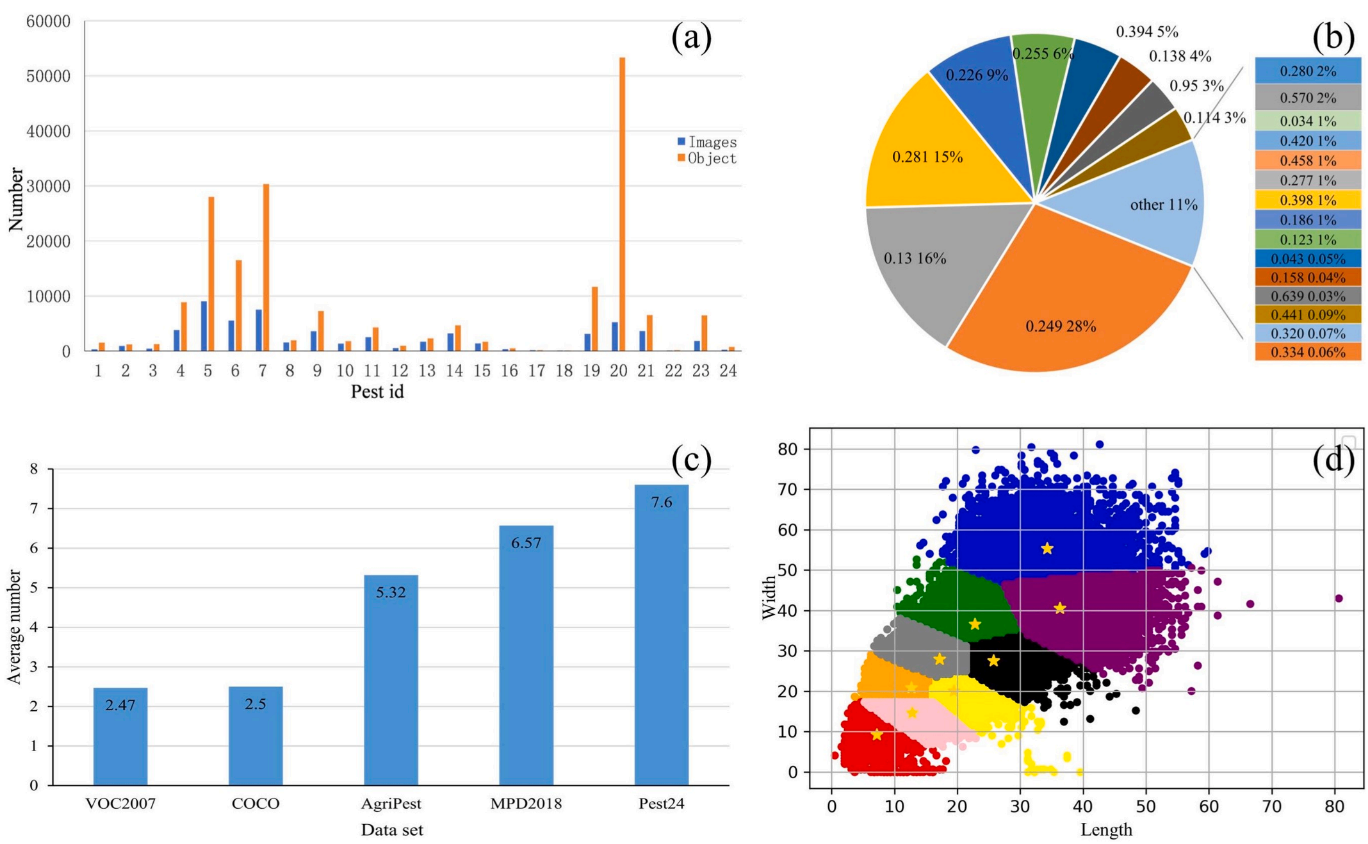

Fig. 2. (a) Distribution of the Pest24 dataset; (b) Relative size distribution of pests; (c) Average number of targets per image in different datasets;(d) Clustering results for the ground truth of the dataset. The horizontal and vertical axes represent the length and width of the ground truth, respectively, and the blue pentagons represent the category centers.

protection experts and agricultural technicians, generating XML files according to the PASCAL VOC standard. The annotation information includes the location coordinates and categories of the pests. The final number of annotated categories, annotated images, and annotated instances are shown in Table 2.

2.2. Data analysis

We have conducted further analysis on the characteristics of the presented image data and have drawn the following conclusions based on our findings.

2.2.1. Class-imbalance

As shown in Fig. 2(a), the number of samples is unevenly distributed across categories. Notably, Anomala corpulenta (20) had a total of 53,347 samples, while only 108 were available for Holotrichia oblita (18). The substantial disparity in sample sizes could have a significant impact on the detection and counting performance of the model and method.

2.2.2. Relatively small size

Relatively small objects are considered when the target size is less than 0.1 times the size of the original image (Y. Li et al., 2020). The report analyzed the relative scale of all labeled pests by calculating the ratio of pixels occupied by the pest label to the total number of pixels in the image. In Fig. 2(b), the different color blocks indicate different scales of the relative size distribution of pests. Specifically, the figure shows the number of pest instances labelled with different proportions. The statistical results, shown in Fig. 2(b), indicate that there were 50,829 instances of labeled samples with a relative scale of 0.249 %. Furthermore,

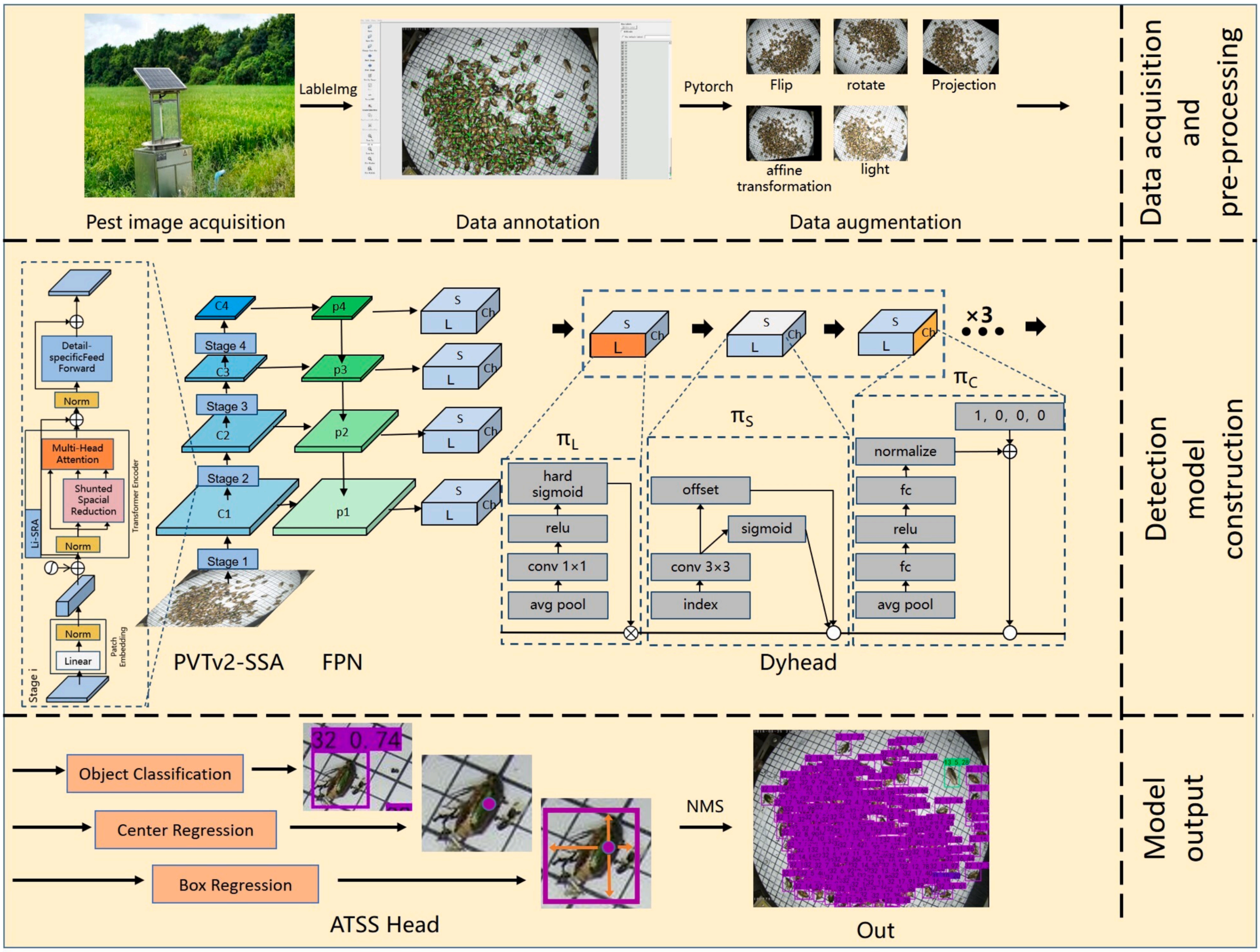

Fig. 3. Flowchart of Pest-PVT Pest Detection Count Framework. Note: The structure of Stage1 to Stage4 in PVTv2 is the same as shown in Stage i, where the Shunted Spacial Reduction module is added to the Transformer Encoder to optimise the K, V vectors. The input image size for the backbone network is 224 × 224, $C1 = \frac{H}{4} \times \frac{W}{4} = 56$, $C2 = \frac{H}{8} \times \frac{W}{8} = 28$, $C3 = \frac{H}{16} \times \frac{W}{16} = 14$, $C4 = \frac{H}{32} \times \frac{W}{32} = 7$. L is the feature map for different Stage, S is the spatial location information ($S = H \times W$), and Ch is the channel information.π_L is scale-aware attention,π_S is spatial-aware attention,π_C is task-aware attention.

there were 28,968 and 27,350 pest annotations with relative scales of 0.13 % and 0.281 % respectively. Most of the pest types within the dataset possess relative scales that are less than 0.4 %, and this leads to an extremely limited quantity of pest features that the detection model can extract.

2.2.3. *Intensive target*

Multiple pests are present in each image, often obscured and stuck together. As shown in Fig. 2(c), the average number of objects per image in the target detection datasets, such as VOC2007, and COCO, is 2.47, and 2.50, respectively. However, pest detection datasets such as AgriPest and MPD2018 have an average of 5.32 and 6.57 pests per image. In contrast, the average number of pests per image in our Pest24 dataset is 7.60.

2.2.4. *Multi-scale target*

In order to illustrate the limitations of anchor-based methods in the Pest24 dataset, we employed the K-means clustering algorithm to cluster the ground truth data and visualize the results. In Fig. 2(d), the colors and shapes represent the K-means clustering results. Here the K-means algorithm was used to cluster the ground truth, producing nine cluster centroids representing anchor boxes with different aspect ratios. In this figure, the category centers are marked by pentagrams in different colored areas, which reflect the size of the anchor boxes obtained through clustering. This algorithm produced nine centroids representing anchor boxes of various sizes with aspect ratios of (8, 9), (12, 15), (12, 21), (17, 28), (19, 20), (22, 37), (26, 28), (37, 41), and (34, 56). It is worth noting that there is a considerable variation in the size of true boxes belonging to the same cluster within the green and grey areas of the figure. The five-pointed stars within the figure indicate the category centers and represent the size of the anchor box obtained through clustering. Clearly, the size of the anchor boxes obtained by clustering does not align well with the size of the true boxes in the Pest24 dataset, making it challenging to train the pest position regression model (Lv et al., 2022).

3. Overall design of the model

The paper presents Pest-PVT, a Transformer-based network model designed for multi-class dense small pest detection and counting tasks. The structure diagram of the proposed pest detection task framework is shown in Fig. 3. The Pest-PVT proposed in this paper uses the ATSS version of FCOS as a baseline model. Based on the anchor-free strategy, FCOS does not rely on the IoU to discriminate positive and negative samples for dense small targets that are more sensitive to position deviation (Zhang et al., 2020). Meanwhile, Adaptive Training Sample

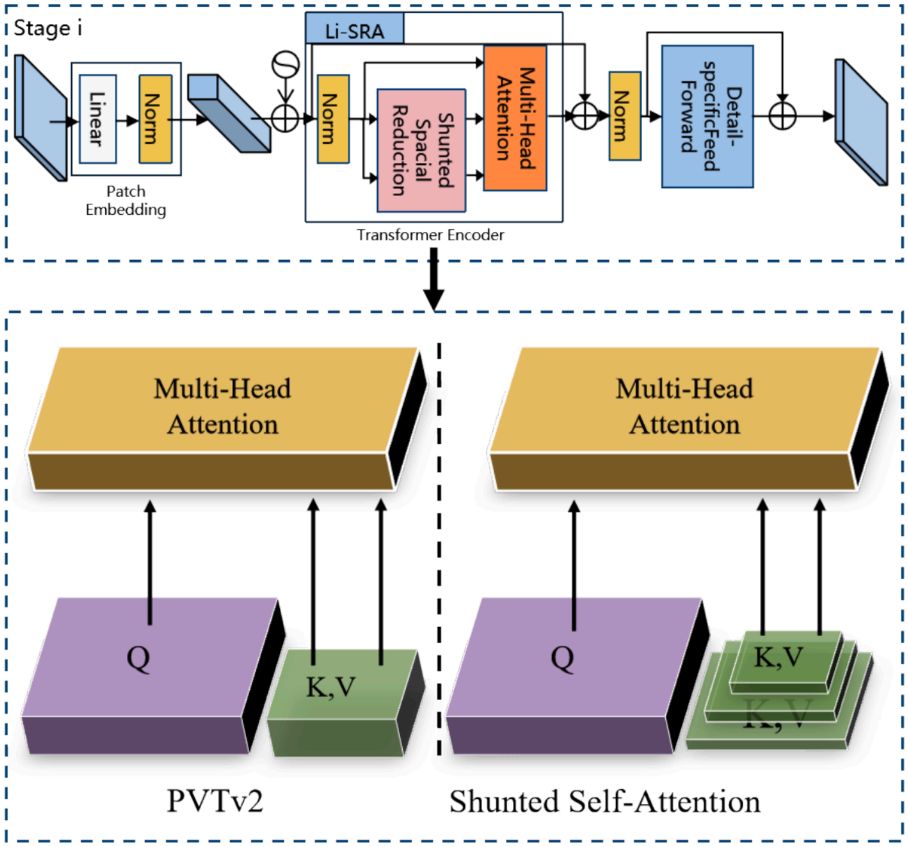

Fig. 4. Comparative chart of multi-headed self-attention mechanisms.

Selection (ATSS) can dynamically select training samples that better match the distribution of pest samples for training. Secondly, the pyramid-structured transformer model PVTv2 is used as the backbone network. To enhance computational efficiency while preserving the ability to capture features for small objects, the network integrates the Shunted Self-Attention (SSA) module (Sucheng Ren et al., 2022).Finally, the stacked DyHead (Dynamic Head) module is used as the neck of the model to perform attention on the level, space and channel dimensions of the feature map output by FPN, thus unifying the three different attentions into one detection task (Dai et al., 2021). The flowchart of the pest detection and counting task framework in this chapter is shown in Fig. 3.

3.1. *Framework of anchor free Adaptive training positive and negative sample Selection algorithm*

Based on the anchor mechanism, object detection algorithms including various variants, such as R-CNN (Girshick et al., 2014); Faster R-CNN (Bochkovskiy, Wang, & Liao, 2020), SSD (W. Liu et al., 2016), RetinaNet (Lin et al., 2017); and YOLO series (Bochkovskiy et al., 2020); Ge, Liu, Wang, Li, & Sun, 2021,C.-Y. Wang, Bochkovskiy, & Liao, 2022; Ge, Liu, Wang, Li, & Sun, 2021), and generate candidate boxes with different scales and aspect ratios. These candidate boxes are used for object position classification and regression, and the best predicted boundary box is selected as the final detection. Although anchor mechanism solves the problem of redundant computation and time–-space consumption in candidate box extraction, it still faces some challenges.

Firstly, the use of rectangular boxes leads to a large amount of background information that interferes with the network's judgement. Secondly, the design of the anchor size and aspect ratio is highly sensitive to the dataset and often requires multiple manual adjustments to achieve satisfactory results (Lv et al., 2022). Finally, the large number of candidate boxes generated by the anchor mechanism leads to an imbalance between positive and negative samples (Zhang et al., 2020). The mainstream anchor-based network, YOLOv4, uses manually set IoU thresholds to select positive samples. That is, if the sample's IoU is greater than the IoU threshold, it is classified as a positive sample. Otherwise, it is classified as a negative sample. This method heavily relies on human intervention, and improper threshold settings can create an imbalance between positive and negative samples. This imbalance can significantly reduce the training effectiveness for specific target classes, thus impacting the overall model training. To address these issues, we chose to use the ATSS. It is capable of obtaining adaptive IoU thresholds based on the characteristics of the overall pest samples in the dataset, without the need for manual adjustment of the model's hyperparameters, to achieve automatic separation of positive and negative samples. During the training process, when an insect image collection G containing g real insect targets is input, first obtain the prior box of the m-th target in the corresponding feature layer. In each feature layer, k candidate positive samples with the smallest distance are selected as candidates using L2 distance. Second, calculate the IoU between these candidate positive samples and the real box, add the mean and variance of the IoU of these candidate positive samples to obtain the final IoU threshold. Thirdly, compare the IoU of the candidate positive samples with the threshold, select the candidate positive samples with IoU greater than the threshold as positive samples, otherwise as negative samples. Repeat the above steps until all real targets are found as positive samples, and finally output the positive samples for network regression training. The ATSS network framework can adaptively determine the threshold based on the input sample. Compared with manually setting the threshold, using this method to obtain positive and negative samples is more reasonable and balanced, which is more helpful to improve the model training effect and detection accuracy.

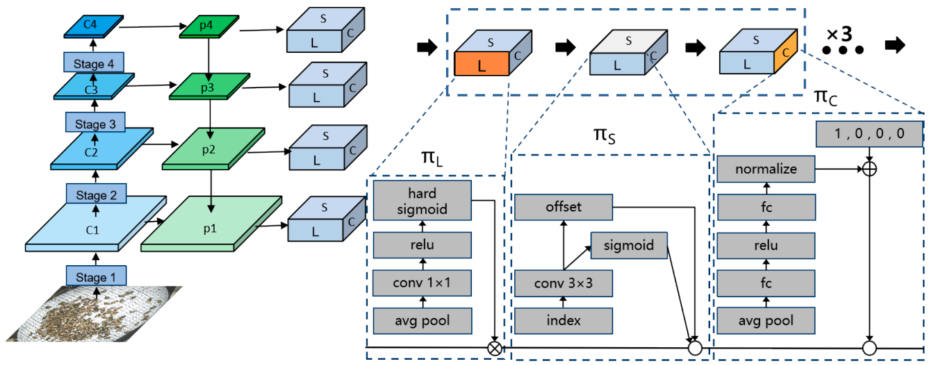

Fig. 5. DyHead structure diagram.

3.2. Optimization of self attention mechanism based on pyramid Vision Transformer V2

CNN has dominated the field of object detection due to its inherent inductive bias, such as translation invariance and locality (Qi et al., 2022). However, convolutional operations can only capture local information and do not model the dependencies between features, limiting their ability to fully exploit contextual information. In addition, CNN weights cannot dynamically adapt to changes in the input. As a result, the development of visual transformer models is beginning to challenge the dominance of CNN.

Inspired by PVTv2, we replaced the CNN block with a PVTv2 encoder block to form the backbone network of our detection model. Compared to convolution, contextual and global information is captured by the transformer encoder module. Self-attention mechanisms are used to exploit the representational potential of information features. In addition, it can effectively handle densely occluded objects.

The self-attention mechanism in the Transformer model results in substantial memory expenses, which increase quadratically with the token count. Therefore, PVTv2 fuses tokens at different stages to obtain different receptive fields for multi-scale information, while keeping the receptive field the same within each layer. By reducing K and V within each layer, a coarser receptive field can be used to reduce computation, but small object details cannot be considered. Considering that pest detection requires the detection of small targets, we improved PVTv2 by incorporating a Shunted Self-Attention (SSA) module into its attention mechanism. This enables the aggregation of multiscale information into the same attention layer and the use of a Shunted Transformer structure to detect multiscale objects (Sucheng Ren et al., 2022).

As shown in Fig. 4, PVTv2 reduces the scale of K and V vectors. Shunted Self-Attention reduces the scale of K and V vectors at different levels and concatenates them along the head dimension. This operation is equivalent to performing multi-level and multi-scale feature extraction and interaction on different parts of the input image, thereby improving the model's expressiveness and generalization ability.

$$Q_i = XW_i^Q$$

$$K_i, V_i = MTA(X, r_i)W_i^K, MTA(X, r_i)W_i^V$$

$$V_i = V_i + LE(V_i) \tag{1}$$

In Equation (1), i represents the number of scales in the K, V dimension. MTA (Multi-scale token aggregation) is a feature aggregation module with downsampling rate r_i, which utilizes depth-wise separable convolutional layers to enhance the connectivity of adjacent pixels in V. W_i^Q, W_i^K, and W_i^V are parameters for linear projections of the i-th attention head. There are different r_i values on attention heads in the first layer. Therefore, K and V can capture different scales in the self-attention mechanism. MTA enhances the local components of the deep convolution. Compared to spatial reduction, it preserves finer and lower-level details. Equation (2) is then used to calculate the sharded self-attention mechanism.

$$h_i = Softmax\left(\frac{Q_i K_i^T}{\sqrt{d_h}}\right)V_i \tag{2}$$

In Equation (2), d_h is the dimension. Due to the multi-scale K and V, shunted self-attention is more powerful in capturing multi-scale targets. The reduction of computational cost depends on the value of r. Therefore, the model and r can be well defined to balance computational cost and model performance. When r increases, more tokens in K and V are merged, the length of K and V becomes shorter, so the computational cost is low, but the ability to capture large targets is still retained. In contrast, when r decreases, more feature details are retained but more computational cost is incurred. By integrating various r into one self-attention layer, multiple downsampling rates of MTA can be obtained, enabling the model to capture multi-grain features of pests (Sucheng Ren et al., 2022).

3.3. Detection head optimization based on multi-dimensional attention mechanism

The purpose of the target detection head is to classify and locate objects. An excellent target detection head should have the abilities of scale perception, spatial perception, and task perception. For the model's scale perception ability, relying solely on features from different levels of the feature pyramid network leads to significant semantic gaps, and directly fusing different feature layers is not optimal (Guo et al., 2020). To address this issue, PANet (S. Liu et al., 2018)proposed to enhance low-level features through a bottom-up path, Libra R-CNN (Pang et al., 2019)introduced balanced sampling and a balanced feature pyramid, and SEPC (X. Wang, Zhang, Yu, Feng, & Zhang, 2020) proposed a pyramid convolution to extract scale and space features.

For task perception ability, various representations of objects can improve performance. FCOS (Tian et al., 2019)proposes to solve the object detection problem pixel-by-pixel using the center representation method. ATSS proposes automatic selection of positive samples to improve accuracy. RepPoints (Yang et al., 2019)simplifies learning by treating object detection as key points. Centernet (Duan et al., 2019) treats each object as three key points instead of one pair to reduce incorrect predictions and improve accuracy. Borderdet (Qiu et al., 2020) extracts boundary features to enhance point features. However, most previous studies aim to improve specific perceptual abilities from different perspectives and fail to propose a unified view. Therefore, this paper uses the dynamic head framework (Dai et al., 2021) to unify the target detection head through attention mechanisms. By seamlessly combining multiple self-attentions in the feature layers of scale perception, spatial positions of spatial perception, and output channels of task perception, the representational ability of the target detection head is significantly improved.

As shown in Fig. 5, the feature map generated by FPN is up-sampled

Fig. 6. NVIDIA Jetson TX2.

or down-sampled to produce a three-dimensional tensor $F \in R^{L\times S\times C}$, where *L* represents the feature pyramid level, *S* represents the height *H* and width *W* of the feature map, and *C* represents the number of channels in the feature map. DyHead employs an independent attention mechanism that connects three different attention mechanisms in sequence, with each mechanism only focusing on one dimension. Among them, π_L, π_S, and π_C are three different attention functions applied to dimensions *L*, *S*, and *C*, respectively. (See Fig 6.).

$$W(F) = \pi_C(\pi_S(\pi_L(F) \bullet F) \bullet F) \bullet F \tag{3}$$

The scale-aware attention mechanism π_L, operates on the level dimension and corresponds to different target scales in the feature maps of different levels. By enhancing the attention at different levels, the scale-aware ability of the target detection can be improved, as shown in Equation (4).

$$\pi_L(F) \bullet F = \sigma\left(f\left(\frac{1}{SC}\sum\nolimits_{S,C} F\right)\right) \bullet F$$

$$\sigma(x) = \max\left(0, \min\left(1, \frac{(x+1)}{2}\right)\right) \tag{4}$$

Where f(x) is a linear function, approximated by a convolution layer with kernel size of 1*x*1, and σ(x) is the hard sigmoid function.

π_S is a spatial-aware attention mechanism that operates on the spatial dimension. Different spatial positions correspond to geometric transformations of the target object. By increasing attention on the spatial dimension, the spatial perception ability of the object detector can be enhanced. Since the *S* dimension is composed of the *H* and *W* dimensions, it is necessary to decouple this module. Firstly, the *S* dimension tensor is sparsified using deformable convolution, and then cross-layer feature aggregation is performed at the same spatial position, as shown in Equation (5).

$$\pi_S(F) \bullet F = \frac{1}{L}\sum_{l=1}^{L}\sum_{k=1}^{K}\omega_{l,k} \bullet F(l; p_k + \Delta p_k; c) \bullet \Delta m_k \tag{5}$$

Among them, *L* represents the feature level, *K* represents the number of positions sampled sparsely, and $\Delta p_k + \Delta m_k$ indicates that position offset is performed to focus on areas with discriminative power. Δm_k represents a self-learned importance measure factor about position p_k. All of these can be learned from the input features at the F intermediate level.

π_C is task-aware attention, which operates in the channel dimension and dynamically switches feature channels to select different tasks. In order to enhance the target detection perception for different tasks we increase the focus on the channel dimension, as shown in Equation (6).

$$\pi_C(F) \bullet F = \max(\alpha^1(F) \bullet F_c + \beta^1(F), \alpha^2(F) \bullet F_c + \beta^2(F)) \bullet F \tag{6}$$

Perform global average pooling in the channel dimension to reduce dimensionality, followed by two fully connected layers and a normalization layer. Finally, a shifted Sigmoid is used to normalize the output. This attention integration module can be stacked according to Equation (3), and in this paper the module has been stacked three times through

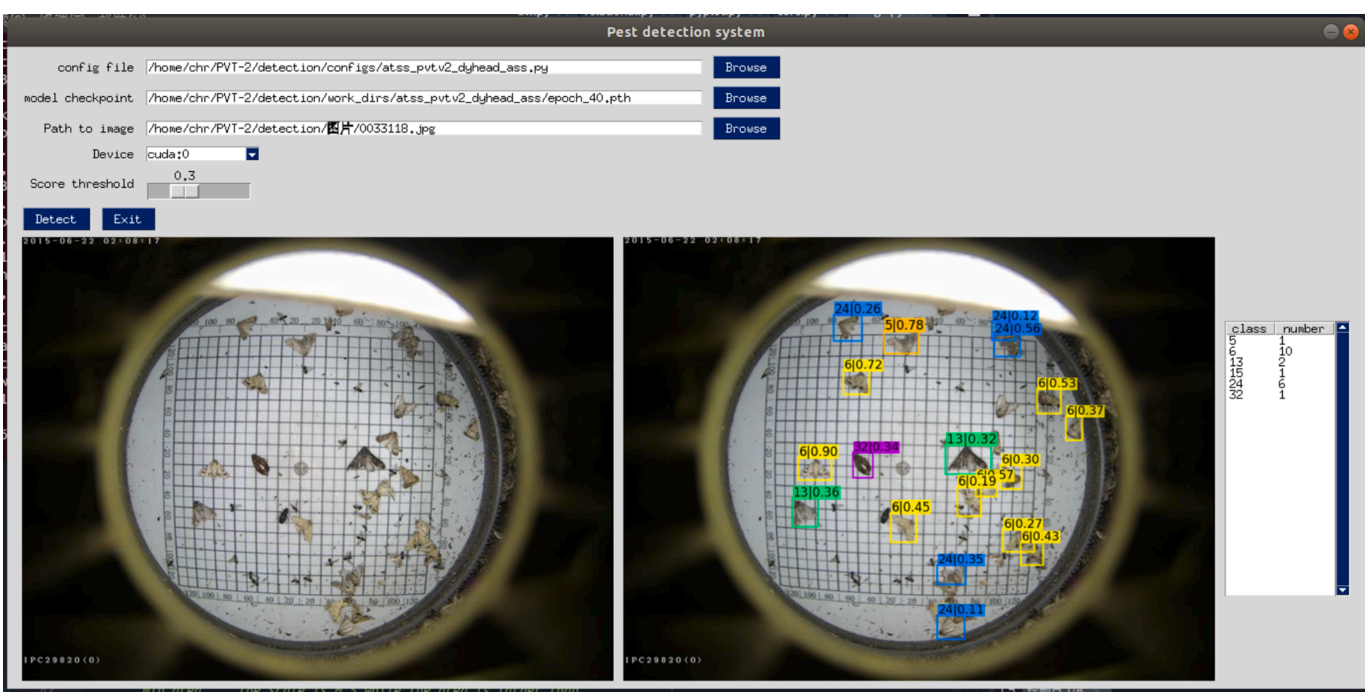

Fig. 7. GUI for Pest Detection System.

Table 3
Experimental hardware and software environment.

Hardware Environment	Equipment	Software Environment	Version
Deep learning server	Dell T7920 server	Operating System	Ubuntu 18.04
CPU	Intel Xeon Silver 4210R@ 2.4 GHz*2	GPU Computing Platform	Cuda 10.0.130
GPU	Nvidia GeForce RTX 2080Ti	Deep Learning Acceleration Library	Cudnn 7.3.1
RAM	SAMSUNG DDR4 16 GB*4	Programming Language Environment	Python 3.7
Embedded AI Computing Platform	Jetson TX2	Model Framework	MMdetection V2.26.0

experiments.

3.4. Embedded integration of algorithms in insect Traps: A Showcase

In order to facilitate the seamless integration of our pest detection and counting model onto a specific model of insect trap device, we conducted preliminary performance testing on an embedded development board. The details of the migration testing are as follows.

3.4.1. Hardware development environment

In this study, we have chosen the NVIDIA Jetson TX2 as the primary hardware device. The NVIDIA Jetson TX2 is an embedded computing platform for artificial intelligence. It features the Tegra X2 processor and Pascal architecture GPU, providing high performance, low power consumption, and strong AI acceleration. It has broad applications in various industries including drones, smart cameras, robotics, semi-autonomous, and fully autonomous driving. With 256 CUDA cores and support for deep learning engine acceleration, Jetson TX2 enables efficient implementation of artificial intelligence applications such as speech recognition, object detection and classification, and natural language processing. Furthermore, Jetson TX2 integrates multiple high-speed interfaces such as Gigabit Ethernet, PCIe, USB 3.0, and HDMI that enable users to perform diverse data transmission and information display tasks. As an embedded computing product, the Jetson TX2 also has the advantage of low power consumption. Its overall power consumption can reach a minimum of 7.5 W, fully meeting the needs of remote control and long-term operation in field environments.

3.4.2. Software development environment

This study utilized NVIDIA's JetPack 3.2 development kit to upgrade the Jetson TX2 development board's firmware, which runs Ubuntu 16.04 operating system, CUDA 9.0, cuDNN 7.0, and Python 3.7. We first built the MMdetection framework on the Jetson TX2 board, then transferred both the Pest-PVT model and trained weight files from the server to the MMdetection framework on the board. Subsequently, a graphical user interface (GUI) was created for the pest detection system using the PySimpleGUI package, as depicted in the accompanying Fig. 7. The GUI features multiple functional modules, such as "config file," which selects the model configuration file, "model checkpoint," which selects the model's weight file, "Path to image," which selects the image to be detected, and "Device," which enables users to run the model using CPU or GPU (set to GPU by default). Additionally, the "score threshold" adjustment module allows for the threshold of the detection box score to vary between 0–1. In the image display area, the left side displays the image to be detected, while the right side displays the detection result. Finally, we conducted counting statistics on the detection results, displaying the number of pests detected for each category in the textbox located on the far right.

4. Experiments

4.1. Experimental setup

The dataset contains a highly unbalanced distribution of pest samples across different categories. Due to a severe imbalance in the number of pest categories within the dataset, we performed data augmentation on the three categories with the fewest sample quantities, namely categories 17, 18, and 22. We combined five common data augmentation methods at random to carry out the enhancement, which included horizontal flips, vertical flips, hue transformations, rotations, and affine transformations. These data augmentation techniques were all implemented using the transformers module within the Pytorch framework. All experiments in this article were conducted in the hardware and software environments shown in Table 3.

We used MMdetection V2.26.0 as the experimental framework, and both the improved model and the comparison model were constructed based on MMdetection. Considering the limitation of GPU memory during training, the batch size was set to 8 and the input image ratio was 224 × 224. Meanwhile, the SGD optimiser was used with an initial learning rate of 0.0001, an impulse of 0.9 and a weight decay of 0.0001. Other settings will be explained in the respective experiments, while ensuring the fairness of the experiments as much as possible.

4.2. Model training

Pest-PVT is used for pest detection and counting tasks. Firstly, all images are resized to 224 × 224 and undergo data augmentation. The preprocessed images are then fed into the PVTv2 backbone network that integrates self-attention mechanism within its SSA module for feature extraction. Secondly, the extracted feature maps from the backbone network are sent to the feature pyramid to fuse features of different scales, resulting in four sizes of feature maps (56 × 56), (28 × 28), (14 × 14), and (7 × 7). For each ground truth box, k candidate boxes with the closest central distance are selected as positive samples separately for each feature pyramid layer, obtaining positive candidate boxes. The IoU value between positive candidate boxes and ground truth boxes is calculated, and the mean m_g and standard deviation v_g of the IoU are computed to obtain the threshold $t_g = m_g + v_g$. Traverse all positive candidate boxes, and if the calculated IoU is greater than the threshold t_g, the candidate box is marked .as a positive sample. Subsequently, the pyramid's output feature maps are sequentially passed through three different attention modules. Finally, NMS operation is performed on the classification and regression network outputs to obtain the final object detection results.

4.3. Evaluation metrics

In this paper, the following metrics were used to evaluate model performance: average precision (AP), Precision, Recall, F1 score, Flops, and Params. They were used to fairly evaluate the performance of the models.

Precision is defined as the fraction of correctly predicted samples among the total predictions predicted as positive (Powers, 2020). The mathematical formula is given by:

$$Precision = \frac{TP}{TP + FP} \tag{7}$$

Recall is defined as the fraction of correctly predicted samples among all the.

predicted samples that are positive in Ground Truth. The mathematical formula is given by:

$$Recall = \frac{TP}{TP + FN} \tag{8}$$

F1-Scoreis the measure of the overall accuracy of the model, based

Table 4
Results of the ablation experiments.

Model	Evaluation indicators					
	Params	FLOPs	mAP	Precision	Recall	F1-score
PVTv2	32.84 M	9.74 GFlops	72.22 %	63.83 %	78.41 %	0.70
PVTv2+①	38.59 M	9.94 GFlops	76.42 %	77.21 %	81.20 %	0.79
PVTv2+②	18.99 M	7.62 GFlops	73.41 %	65.21 %	79.21 %	0.72
Pest-PVT (our)	24.74 M	7.82 GFlops	77.24 %	78.42 %	81.27 %	0.80

① represents the Dyhead module (Dynamic Head), and ② represents the improved self-attention mechanism SSA (Shunted Self-Attention).

on the precision and recall, i.e., the harmonic mean of Precision and Recall (Powers, 2020). The mathematical formula is given by:

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{9}$$

Average Precision is given by the arithmetic mean of precision scores for each sample retrieved. It essentially summarizes the precision-recall curve in a value (Powers, 2020). The mathematical formula is given by:

$$AP = \int_0^1 P(R)dR \tag{10}$$

Furthermore, FLOPs and Params are used to measure the computational performance of the method. The former indicates the model computational power, and the latter represents the number of parameters.

Pest count indicators include Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). MAE is the average difference between actual and predicted pest numbers across all samples and represents the prediction accuracy for each sample. A lower MAE indicates a more accurate model. RMSE is the square root of the mean squared difference between actual and predicted pest numbers across all samples. Unlike the MAE, the RMSE pays more attention to cases with large errors, indicating greater model robustness. A smaller RMSE value represents a more accurate prediction of pest numbers. The Equation for calculations are presented in Equation (11) and Equation (12).

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \widehat{y}_i| \tag{11}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}|y_i - \widehat{y}_i|^2} \tag{12}$$

Table 5
Comparative experimental results of different detection frameworks.

Method	Backbone	Evaluation indicators					
		Params	FLOPs	mAP	Recall	Precision	F1-score
Cascade R-CNN	PVTv2	77.77 M	210.08GFlops	71.0 %	76.33 %	63.94 %	0.70
RetinaNet	PVTv2	45.88 M	71.31GFlops	68.46 %	72.40 %	64.81 %	0.68
GFL	PVTv2	33.11 M	10.03GFlops	70.46 %	77.61 %	60.78 %	0.68
Sparse R-CNN	PVTv2	106.91 M	14.39GFlops	69.96 %	73.83 %	65.81 %	0.70
ATSS(FCOS)	PVTv2	32.84 M	9.74 GFlops	72.22 %	63.83 %	78.41 %	0.70
ATSS(FCOS)	**PVTv2+ SSA**	**18.99 M**	**7.62 GFlops**	**73.41 %**	**65.21 %**	**79.21 %**	**0.72**

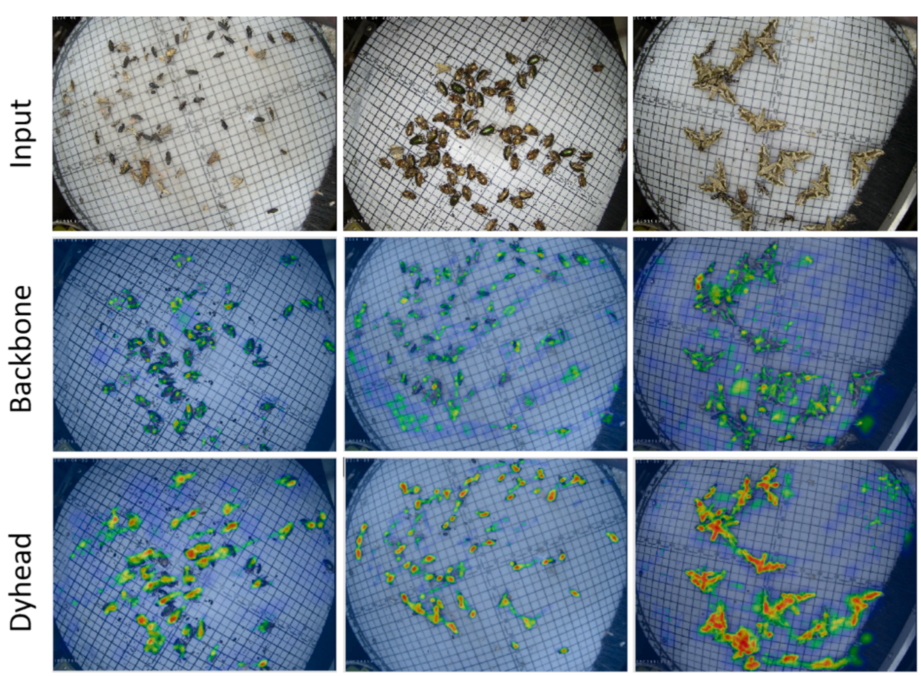

Fig. 8. Heat map of attention distribution.

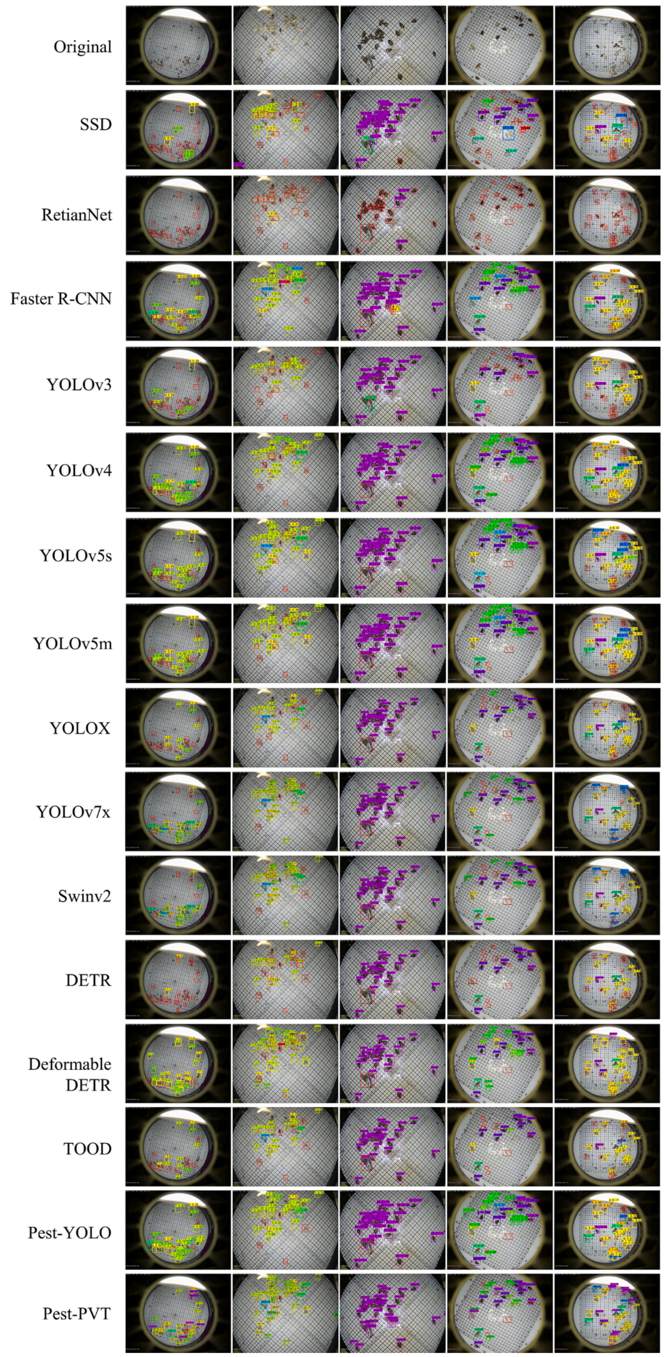

Fig. 9. Detection results for each model.

where N is the number of images, i is the first image, y_i is the number of annotations for the i-th image, and $\hat{y}_i$ is the number of predictions for the i-th image.

5. Results

5.1. Ablation experiment

To verify the effectiveness and necessity of our improved model, we conducted ablation experiments on the optimization of the dynamic head and multi-head attention mechanism of the backbone network. The experimental results are shown in the Table 4.

As shown in Table 4, the introduced Dyhead module in PVTv2+①(Dynamic Head) increases both parameter quantity and computational complexity compared with the baseline model PVTv2. However, the PVTv2+①(Dynamic Head) model achieves better results in other evaluation indicators, with an increase of 4.2 % in mAP, 13.38 % in Precision, 2.79 % in Recall, and 8.78 % in F1-score. Due to the improvement of the self-attention mechanism in PVTv2 by introducing the SSA algorithm, the PVTv2+② (SSA) model reduces the parameter quantity to only 57.8 % of the original model, which is only 18.99 M. At the same time, the computational complexity also decreases by 2.12 %. Despite this, compared with the baseline model, other evaluation indicators have only slightly improved.

Finally, our complete improved model, Pest-PVT, reduces the parameter quantity to only 75.3 % of the baseline network PVTv2, which is only 24.74 M, while reducing the computational complexity by 1.92GFlops. All other indicators have significantly improved, with an increase of 5.02 % in mAP, 14.59 in Precision, 2.86 % in Recall, and 9.45 % in F1-score. Therefore, by introducing Dyhead and improving the self-attention mechanism with SSA, the performance of the model in pest detection tasks can be improved. Given the available resources, choosing the Pest-PVT model with moderate parameter quantity and better performance in evaluation indicators would be a better choice.

To verify the necessity of using the ATSS version of FCOS as a baseline model, we juxtaposed a number of baseline models like Cascade R-CNN (Cai and Vasconcelos, 2018), RetinaNet (Lin et al., 2017), GFL (Hu et al., 2020), and Sparse R-CNN (Sun et al., 2021)as norms. The backbone networks in these baseline models are all PVTv2. To make the comparison fairer, we ensured that all configurations were the same, including input image size, pre-training, and fine-tuning. The parameter quantity of the ATSS version of FCOS is the smallest among the compared detector frameworks, only 32.84 M. However, other evaluation indicators, including mAP, Recall, and Precision, are the highest among the comparison Table 5.

For the dynamic object detection head with scale-awareness, spatial awareness, and task awareness, we visualized its attention distribution, as shown in Fig. 8.

We selected three images of pest instances with small, medium, and large sizes for visualization, as shown in Fig. 9. From the figure, it can be seen that the model with Dyhead pays more attention to the pest instance area. The Dyhead performs scale-awareness attention mechanism in the Level dimension, which enhances the feature extraction ability for pests of different scales. In addition, deformable convolutions are used in the spatial-awareness attention mechanism, making it more adaptable to the shape of the pests. Finally, Dyhead deploys task-awareness attention mechanism to the channel dimension, which can better support anchor-free one-stage detectors in an adaptive manner. The above-mentioned attention mechanisms play an important role in pest detection tasks.

5.2. Comparative experiments

We compared our proposed Pest-PVT network with other advanced detectors on a pest dataset, including SSD (W. Liu et al., 2016), Retinanet (Lin et al., 2017), YOLOv3(Redmon and Farhadi, 2018), YOLOv4 (Bochkovskiy et al., 2020), YOLOv5s, YOLOv5m, YOLOX (Ge et al., 2021), YOLOv7-X (C.-Y. Wang, Bochkovskiy, & Liao, 2022), DETR (Carion et al., 2020), Deformable DETR (Zhu et al., 2020); TOOD (Feng et al., 2021); Swinv2(Liu et al., 2022), and Faster R-CNN (Shaoqing (Ren et al., 2015). We also compared the pest detection models of related work, including AF-RCNN (Jiao et al., 2020); YOLOv3-W (Q.-J. (Wang et al., 2020), YOLOv3-GCNet (Lv et al., 2022), Pest-YOLO (Tang et al., 2021), Faster R-CNN-TSD (Huang et al., 2022), Pest Deformable DETR (Qi et al., 2022), BF-YOLO (Qi et al., 2023), YOLOv5-PY1 (Teixeira et al., 2023), and Pest-YOLO (Wen et al., 2022). The experimental results are shown in the Table 6.

The mAP, Precision, and F1-Score of the proposed Pest-PVT in this paper are the highest in the Table 6, reaching 77.2 %, 78.42 %, and 0.80. Computational complexity of proposed Pest-PVT in this paper is only 12

Table 6
Comparison results of the proposed model and several state-of-the-art detectors.

Model	Params	Flops	mAP	Recall	Precision	F1-score
Faster Rcnn (Ren et al., 2015)	41.53 M	46.89G	42.67 %	54.00 %	45.58 %	0.49
SSD (W. Liu et al., 2016)	31.85 M	64.36G	25.06 %	47.06 %	14.48 %	0.22
RetinaNet (Lin et al., 2017)	57.12 M	62.59G	26.11 %	11.22 %	68.93 %	0.19
YOLOv3 (Redmon and Farhadi, 2018)	38.53	19.38G	60.69 %	44.44 %	59.20 %	0.51
YOLOv4 (Bochkovskiy, Wang, & Liao, 2020)	64.06 M	29.95G	64.27 %	49.59 %	81.99 %	0.62
YOLOv5s	**7.02 M**	**1.58G**	65.54 %	64.26 %	64.38 %	0.64
YOLOv5m	20.9 M	4.8G	66.89 %	70.90 %	64.84 %	0.68
YOLOX (Ge, Liu, Wang, Li, & Sun, 2021)	54.17 M	32.83G	68.88 %	54.58 %	65.59 %	0.60
YOLOv7-x (C.-Y. Wang, Bochkovskiy, & Liao, 2022)	70 M	189.9G	72.92 %	70.51 %	72.73 %	0.69
DETR (Carion et al., 2020)	41 M	86G	37.84 %	71.82 %	32.18 %	0.44
Deformable DETR (Zhu et al., 2020)	40 M	173G	60.72 %	46.24 %	58.40 %	0.52
TOOD (Feng et al., 2021)	31.98 M	201G	68.36 %	75.02 %	60.89 %	0.67
Swinv2(ATSS) (Liu et al., 2022)	42 M	5.46G	68.65 %	73.01 %	49.78 %	0.59
YOLOv3-W (Q.-J. (Wang et al., 2020)	/	/	63.57 %	/	/	/
AF-RCNN (Jiao et al., 2020)	/	/	56.42 %	**85.10 %**	/	/
Pest-YOLO (Tang et al., 2021)	/	/	71.6 %	83.5 %	/	/
YOLOv3-GCNet (Lv et al., 2022)	/	/	65.69 %	/	/	/
Faster R-CNN-TSD (Huang et al., 2022)	/	/	51.0	/	/	/
PestDeformable DETR (Qi et al., 2022)	40 M	170G	72.5 %	/	/	/
BF-YOLO (Qi et al., 2023)	/	66.3G	74.1 %	/	/	/
YOLOv5-PY1 (Teixeira et al., 2023)	/	/	72.1 %	70.7 %	69.4 %	/
Pest-YOLO (Wen et al., 2022)	64.04 M	30.09G	69.59 %	77.71 %	46.94 %	0.53
Pest-PVT (our)	24.74 M	7.82G	**77.20 %**	81.27 %	**78.42 %**	**0.80**

% of BF-YOLO. respectively, while the recall rate is 81.27 %, second only to AF-RCNN, which is specifically optimized for pest recall rate. Compared with the most advanced detector in the current YOLO series, YOLOv7-x, Pest-PVT's mAP, recall, and precision are higher by 4.28 %, 10.76 %, and 5.69 %, respectively. Compared with Swinv2, a backbone network that also uses transformers, Pest-PVT has the highest evaluation indicators, including mAP, recall, and precision. Compared with the related work Pest-YOLO (Wen et al., 2022), Pest-PVT's mAP, recall, and precision are higher by 7.61 %, 3.56 %, and 31.48 %, respectively.

At the same time, we also conducted statistics on the model's parameter size and computational complexity. The Params and FLOPs of Pest-PVT proposed by us are much lower than YOLOv7-x. Its Params is 2.8 times smaller than YOLOv7-x, and its FLOPs is reduced by 24 times compared to YOLOv7-x. Although Params and FLOPs are not the lowest among all compared models, they are lower than most models while achieving higher evaluation indicators. Overall, Pest-PVT has a better balance between accuracy and performance, achieving higher detection accuracy with a smaller parameter model and lower computational complexity, making it more suitable for integration into outdoor field pest detection devices with limited hardware platforms.

In order to have a more intuitive understanding of the detection performance of the Pest-PVT model, we compared the detection results of the model through visualization. Among them, AF-RCNN (Jiao et al., 2020), YOLOv3-W (Q.-J. Wang et al., 2020), YOLOv3-GCNet (Lv et al., 2022), Pest-YOLO (Tang et al., 2021), Faster R-CNN-TSD (Tang et al., 2021), Pest Deformable DETR (Qi et al., 2022), BF-Y OLO (Qi et al., 2023), and YOLOv5-PY1(Teixeira et al., 2023) were not included in the comparison as the authors did not open-source their models. The detection results are shown in the Fig. 9, where we selected five pest images for detection and displayed their visualization results. In order to show the pest detection of different models more clearly, we marked the pests missed and misdetected by the models with red borders, and the other colors of the borders indicate the different categories of pests detected by the models. In the five test images shown in the Fig. 9, mainstream CNN target detection models such as YOLOv7x and TOOD have 20 and 31 pests misdetected, respectively. Representative Transformer models, such as Swinv2 and DETR have 20 and 53 pests misdetected, respectively. The Pest-PVT proposed in this paper has only 7 pests misdetected and performs much better in real pest detection tasks.

5.3. Counting experiment

To test the counting ability of the model, this paper selected 50 images with multiple, dense and occluded attached individuals from the test set to test the performance of the model in counting the number of pests. In this paper, the count regression curves and count error plots of each model are plotted, and the R2, *MAE* and *RMSE* of different model counts are compared.

To avoid the problem of label loss affecting the counting experimental results, this paper tested the true counting ability of the model by

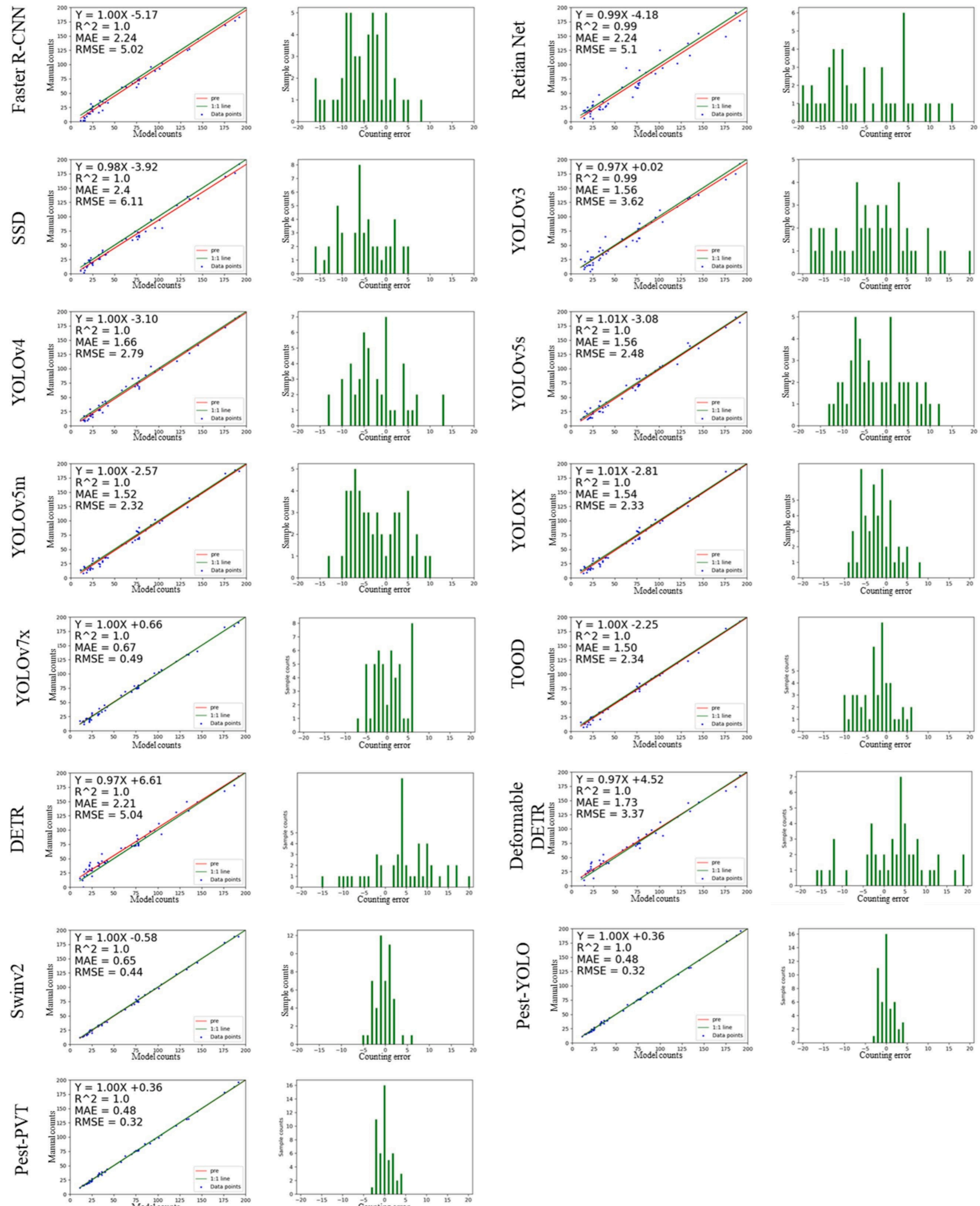

Fig. 10. Plot of the counting test results of the models. Note: The left subplot is a linear regression plot between the model-predicted and manually counted pest numbers, where the red line represents the regression curve of the model prediction and the green line represents the actual pest count. The right subplot is a histogram of counting errors, where the x-axis represents the counting errors of the model and the y-axis represents the number of samples.

manually counting the 50 selected images. The manual counting results of these 50 images showed a total of 3855 pests. On the re-labeled 50 images, Pest-PVT, Pest-YOLO, TOOD, DETR, Deformable DETR, YOLOX, YOLOv5m, YOLOv5s, YOLOv4, YOLOv3, SSD, Faster R-CNN, RetinaNet, YOLOv7x and Swinv2 detected 3861, 3844, 3823, 3788, 3865, 3613, 3836, 3740, 3738, 3717, 3734, 3567, 3604, 3594, and 3842 pests, respectively. After statistical analysis, the detection results of Pest-PVT were closest to the number of re-labeled pests, with a counting error of only 6 pests, higher than the 13 pests of YOLOv7x and the 11 pests of Pest-YOLO. This indicates that Pest-PVT method is more accurate in counting pests than other comparative models.

We conducted statistical analysis on the counting results of 14 detection models and the proposed Pest-PVT model, which were visually represented in Fig. 10. For each detector, two visualizations were presented. The left subplot in Fig. 10 is a linear regression plot between the model-predicted counting results and manual counting, while the right subplot is a histogram of counting errors. From the linear regression plot, it can be observed that the slope of the regression line is 1 and the intercept is only 0.36, indicating a good fit between the linear regression line (in red) of the Pest-PVT counting result and the ground truth curve (in green). The RMSE of the Pest-PVT model was 0.32, which is lower than that of other detectors in the figure, suggesting that the Pest-PVT model has better counting stability than other models. Based on the histogram of counting errors, the counting errors of the Pest-PVT model is concentrated in the range of −4 to 4. Specifically, the Pest-YOLO model is concentrated within that range, YOLOv7x is concentrated in the range of −7 to 6, Swinv2 is concentrated in the range of −7 to 8, YOLOX is concentrated within that range, while TOOD is concentrated in the range of −10 to 7. This indicates that the counting error of the Pest-PVT model is smaller than that of other models.

6. Conclusions

This paper proposes an improved Pest-PVT detection model based on the Pyramid Vision Transformer v2. Firstly, in this paper, the ATSS version of FCOS is used as the anchor-free baseline network to avoid the effect of inaccurate detection box positioning due to preset anchor boxes. In addition, ATSS can dynamically select training samples that better match the distribution of pest samples for training purposes. Second, the feature extraction network is Pyramid Vision Transformer v2, which is known for its strong feature pyramid structure and feature extraction ability, and its good fitting effect for large datasets. To address the high computational cost of the self-attention mechanism in Pyramid Vision Transformer v2, the study integrates SSA (Shunted Self-Attention) into the attention mechanism to improve the computational efficiency of the feature extraction network while maintaining the feature extraction ability for small targets. Finally, the study uses a stacked DyHead (Dynamic Head) as the model head, which unifies the three different attentions into the detection model, significantly improving the performance of the detection model without increasing the computational cost. In the experiment, the mAP reached 77.24 %, which is higher than other compared models, and the parameter volume and computational complexity are only 24.74 M and 7.82GFlops, making it more suitable for use on field pest detection hardware devices with limited resources.

Despite some of the successes achieved in this work, future work should continue to focus on the optimisation of pest detection models. A key area for further improvement lies in the parameter size and inference speed of the model. While Pest PVT is relatively efficient, we need to continue to explore more efficient pruning and compression of the model without sacrificing performance, which is critical for wider adoption in resource-constrained agricultural environments. In future work, we can use strategies such as neural architecture search and knowledge distillation to design a model that can be adapted to lower power and computationally constrained edge devices. Addressing labelling errors and missing labels in datasets is another pressing issue. To mitigate the larger impact of labelling errors and missing labels on fully supervised training. We plan to use semi-supervised or weakly supervised learning methods to learn from noisy labels more efficiently, while also improving the robustness and generalisation of the model to make it resilient to inconsistent data annotations. Finally, in the future, we plan to integrate the set of work at the edge to tailor the model for specific edge devices, such as Pest trapping equipment or unmanned aerial vehicle. The combination of hardware acceleration strategies and model speed optimisation will ensure efficient response of real-time pest detection models in the field.

CRediT authorship contribution statement

Hongrui Chen: Writing – original draft, Methodology, Investigation, Data curation. **Changji Wen:** Writing – review & editing, Supervision, Investigation. **Long Zhang:** Validation, Software. **Zhenyu Ma:** Resources, Investigation. **Tianyu Liu:** Software, Resources. **Guangyao Wang:** Supervision, Software, Resources. **Helong Yu:** Validation, Resources. **Ce Yang:** Writing – review & editing, Validation, Supervision, Resources. **Xiaohui Yuan:** Writing – review & editing. **Junfeng Ren:** Visualization, Validation, Software, Formal analysis.

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Declaration of competing interest

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Data availability

The data that has been used is confidential.

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